Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Smith, Carolyn L Ph.D.		
Project Title:	Receptor Countermeasures to Bone Loss in Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	05/31/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Correct end date is 5/31/2006, per K. Major/NSBRI (12/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Weigel, Nancy (Baylor College of Medicine) Suva, Larry (University of Arkansas College of Medicine) Peterson, Leif (Baylor College of Medicine) Luo, Zong-Ping (Baylor College of Medicine)		
Grant/Contract No.:	NCC 9-58-BL00407		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The prevention of bone loss due to skeletal unloading is a complex problem and the reasons for this loss have not been elucidated. The overall goal of the bone team of NSBRI is to develop countermeasures that will not only prevent quantitive loss of bone, but also maintain bone strength. Measures that simply prevent resorption may maintain mass, but block the necessary remodeling to ensure appropriate bone strength. Biological actions mediated by the ER and VDR play key roles in the normal control of bone growth and turnover that is necessary for skeletal health. These receptors act by controlling (1) the differentiation of the precursors of OBs and OCs, (2) the functions, including gene expression, of mature OBs and OCs and/or (3) other cell types within the bone and bone marrow microenvironment that control (1) and (2) (e.g. stromal cells and monocytes). In addition, they play important roles in regulating the expression of systemic factors that contribute to the overall balance of bone remodeling in the skeleton. Finally, VDR is critical for efficient intestinal calcium absorption. We hypothesize that the appropriate combination of an agent that will improve calcium absorption (VDR agonist) and an agent that will reduce bone resorption (a bone-selective estrogen receptor agonist, STEAR) will achieve these goals. Indeed, there may be synergistic interactions between these receptors that enhance the actions over any one used alone. This proposal outlines studies designed to investigate the physiological, cellular and molecular alterations that occur in the hindlimb suspension (HLS) rat model of skeletal unloading and to evaluate the ability of ER and VDR agonists to counteract the deleterious effects of mechanical unloading on the mature rat, with respect to parameters associated with the skeleton and its associated soft tissues, endocrine system and calcium homeostasis. In so doing, we will obtain molecular, cellular and tissue data that will provide information on the ability of our countermeasures to prevent unloading-induced bone loss, as well as information on the mechanisms that contribute to this undesired consequence of microgravity. In this grant's two specific aims, we are examining the role of these agonists in both male and female animals. This is important since astronaut crews are composed of both genders, and previous unloading studies have revealed gender differences in unloading-induced bone loss. This will lead to a better understanding of the causes of skeletal unloading and the means to prevent bone loss. In addition, the results of these studies will provide a foundation for future experiments that will examine the ability of the most promising pharmacological countermeasure(s) to maintain bone mineral density and strength in established models of unloading-induced osteopenia in humans and/or during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The applications of this research to Earth-based populations are most relevant to populations for whom osteoporosis is a concern. For instance individuals at risk for developing postmenopausal osteoporosis would benefit from the development of protocols able to prevent or treat this condition that is associated with a significant morbidity and high economic cost (approximately $17 billion according for the National Osteoporosis Foundation) for the 10 million Americans estimated to have this disease. It is clear that estradiol is no longer a viable option for prevention of osteoporosis for most women, and the raloxifene is not well tolerated by a significant number of women. The ability to use EB1089 represents an opportunity to employ a drug that affects the activity of a relatively unexplored pharmacological target relative to osteoporosis, the vitamin D receptor. The ability of EB1089 to exert a greater effect in females than males, suggests that this agent may be particularly effective in the treatment or prevention of postmenopausal osteoporosis. Moreover, it should be noted that administration of vitamin D receptor ligands is currently being evaluated for treatment and/or prevention of various types of cancer, such as breast and prostate, and use of EB1089 for osteoporosis has the potential to impact the incidence and outcome of diseases that affect hundreds of thousands of Americans each year. Finally, although it is not the intent of this project, nor the mission of NSBRI, to determine the molecular mechanisms by which gender-specific actions of EB1089 are achieved, the data produced in this application has provided important preliminary data that we are pursuing though other venues. Ultimately, this may provide information on the differences between males and females that can be exploited to develop compounds for the prevention of disuse-induced osteoporosis with optimal efficacy for males and females.
Task Progress:	The progress on the tasks outlined in our initial proposal has been as expected. In the past year we have completed the experiments (e.g. measurements of serum markers, etc) on the female loaded and hindlimb-unloaded rats funded in the last granting cycle. This data is now being compiled and a manuscript is in preparation. Based on the findings of the last study, we undertook a study in the current year in which we directly compared 1,25-D and EB1089 for their effects in male versus female rats. This was important for two reasons. First, it will enable us to draw firmer conclusions about gender dimorphic responses since all animals will be examined at the same time. The study is more comprehensive in terms of parameters to be assessed, and this should provide more robust data that may help to pinpoint the basis for the differences in responses. Secondly, we planned in Aim 2 of our grant proposal to evaluate the ability of the STEAR, tibolone, in combination with a VDR ligand to serve as a countermeasure for unloading-induced bone loss in males and females. Based on the results we now have in hand, it is clear that the Aim 2 studies should employ different VDR ligands depending on whether we are investigating bone loss in male or female rats. Our Aim 1 studies are well underway. In the past year, we have built the cages necessary for the hindlimb unloading studies, trained personal at our institution in the use of this model, and have nearly completed the direct experimentation for the males for all the outlined Aim 1 studies. The female rats to be used for Aim 1 studies have been purchased, and are currently aging at our institution while being fed the required calcium/phosphorus/vitamin D diet. Experiments on these animals should be complete by early Fall 2005. We have purchased equipment for mechanical testing of soft tissues (e.g. ligaments), and pilot experiments for this analysis will be conducted this summer, to be followed by analyses of tissues collected from the experimental animals. Microarray experiments will begin this summer, as planned. In addition to the laboratory work, we have submitted 6 abstracts in the past year, resulting in poster presentations at six national meetings.
Bibliography Type:	Description: (Last Updated: 02/09/2006)
Presentation	81. Karmakar S, Narayanan R, Perrien DS, Suva LJ, Weigel NL and Smith CL "Gender- and Tissue-Specific Effects of 1,25-Dihydroxyvitamin D and EB1089 in Intestine and Bone of Mature Rats. " Sep-2005
Presentation	Hogan, H. A., W. E. Rogers, MR Allen, SA Bloomfield and CL Smith "Mechanical properties of bone in hindlimb unloaded adult male rats treated with raloxifene and estradiol. " Jan-2005
Presentation	Karmakar S, R Narayanan, DS Perrien, L Suva, NL Weigel and CL Smith "1,25-Dihydroxyvitamin D3 and its analogue EB1089 show a gender-specific skeletal response in mature ambulatory rats. " Jun-2005

Presentation	Liu, X., CY Lee, CL Smith and ZP Luo "Time dependence of the combined effects of estrogen and cyclic mechanical loading on ACL fibroblast gene expressions " Feb-2005
Presentation	Narayanan R, Karmakar S, Seng C, He Q, Suva LJ, Weigel NL, Bloomfield SA and Smith CL "Evaluation of Estrogen and Vitamin D Receptor Ligands as Countermeasures for Disuse-Induced Bone Loss in Unloaded and Ambulatory Rats: Are responses to Vitamin D Receptor Ligands Gender-Specific? " Jan-2005
Presentation	Narayanan R, Seng C, Weigel NL, Bloomfield SA and Smith CL "Attentuation of Disuse-Induced Bone Loss by Estrogen and Vitamin D Receptor Ligands in Unloaded and Ambulatory Female Rats. " Oct-2004